

PAM131440Hz8DCValenceRun3

February 25, 2025

1 Standard Maxx Analysis Template

1.1 Notes

Overview: See below. Note that this run repeated an earlier one but 8% duty cycle instead of 32% (at 40Hz).

Rationale: PAM13.14 driver activation increased number of events without affecting during so we tested for general valence.

Date of Experiment: 2/19/25

Genotypes: PAM13.14>Chrimson and PAM02>wCS

Sex: Females

Light Paradigm: 10 off; 60min on; 10 off; 40Hz, 8% DC, 18V

Design: Two genotypes alternating in groups of three, beginning with Chrimson on the left and Control on the right

Misc: This was the first run of the day, and it started around 3:00pm. There were not enough flies to complete all the regions.

1.2 Configuration

Configuration was standard. No new or untested code.

```
[1]: #!/autoreload 1
import warnings
from Arena import *
import Tracker
import TwoChoiceTracker
from ExperimentalDesign import *
from Parameters import *

# Suppress all warnings
warnings.filterwarnings('ignore')
```

1.3 Parameters

Just a standard arena max, accepting the default values and a Two-Choice Tracker.

```
[2]: p=Parameters()
p.set_arena_max_values(TrackingType.TWOCHOICETRACKER)
p.print()
```

```

tracking_type: TrackingType.TWOCHOICETRACKER
fps: 0
mm_per_pixel: 0.145
speed_window_seconds: 1
micromove_speed_mm_sec: [0.2, 2]
walking_speed_mm_sec: 2
sleep_threshold_min: 5
interaction_distances: [8]

```

1.4 Read in the data

Next we will read in the data to an Arena object. The format is `Arena(ExperimentName, parameters, data_path=".")`. This will automatically calculate all relevant metrics for the tracking type in question.

The name of the experiment and predefined parameters object are required, but `data_path` is optional (if no data directory is specified, it defaults to the current working directory).

The following files must be present in the defined data directory : * The DTrack experiment file, which is an excel sheet with the name '*ExperimentName.xlsx*'. * The DTrack data files themselves, which are named '*ExperimentName_Data_1.csv*', '*ExperimentName_Data_2.csv*', etc. * An experiment file (most of the time, definitely for a TwoChoiceTracker), which is named '*ExperimentName_Design.txt*'. Note that you can incorporate flipped axes to orient lights for data visualization.

```

[3]: arena = Arena.Arena(p,data_path='./Data/')
      arena.print_short_data_quality_report()

```

	Tracker	HighQuality	NotFound	Indiscernible	StartMinutes	EndMinutes
0	T_0_0	0.833380	0.166620	0.0	0.005294	80.016285
1	T_1_0	0.999555	0.000445	0.0	0.005294	80.016285
2	T_2_0	0.999778	0.000222	0.0	0.005294	80.016285
3	T_3_0	0.999778	0.000222	0.0	0.005294	80.016285
4	T_6_0	0.999833	0.000167	0.0	0.005294	80.016285
5	T_7_0	0.999833	0.000167	0.0	0.005294	80.016285
6	T_18_0	0.920206	0.079794	0.0	0.005294	80.016285
7	T_21_0	0.998999	0.001001	0.0	0.005294	80.016285
8	T_22_0	0.998666	0.001334	0.0	0.005294	80.016285
9	T_23_0	0.997582	0.002418	0.0	0.005294	80.016285
10	T_27_0	0.999833	0.000167	0.0	0.005294	80.016285

Warning: Some data quality falls below 0.9!

	Tracker	HighQuality	NotFound	Indiscernible	StartMinutes	EndMinutes
0	T_0_0	0.83338	0.16662	0.0	0.005294	80.016285

1.5 Summary

Get the summary results for all trackers, including treatments, etc.

This will copy the results to the clipboard and save the data in a file called '*Experiment-*

Name_Summary.csv.

The semicolon at the end of the command will suppress the output to this notebook. Remove it if you want to see the results dataframe.

```
[4]: arena.summarize(copy_to_clipboard=True,write_to_csvfile=True);
```

Or you can also create a faceted summary if you want to separate the different phases of the experiment.

The file name will be '*ExperimentName_Summary_Facet.csv*'.

```
[5]: arena.  
      ↪ summarize_facet(cutoffs=(10,70),copy_to_clipboard=True,write_to_csvfile=True);  
      ↪
```

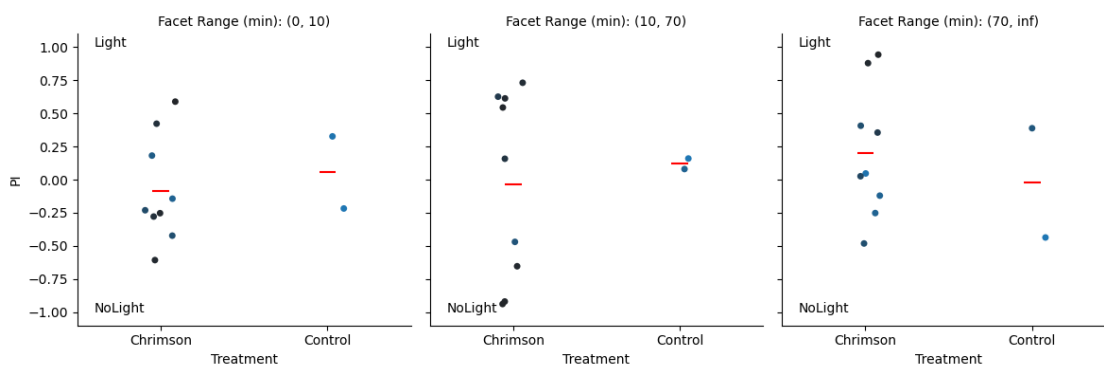
1.6 Plotting

There are lots of cool plotting funtions. Some are shown below.

Facet plots: First of PI, then Percentage, then Total Distance (in mm/min).

Cutoffs separate the acclimaton phase from the experiment phase from the cool down phase.

```
[6]: arena.plot_pi_facet(cutoffs=(10,70))
```



```
[7]: arena.run_pairwise_comparisons_facet(metric="FinalPI",cutoffs=(10,70))
```

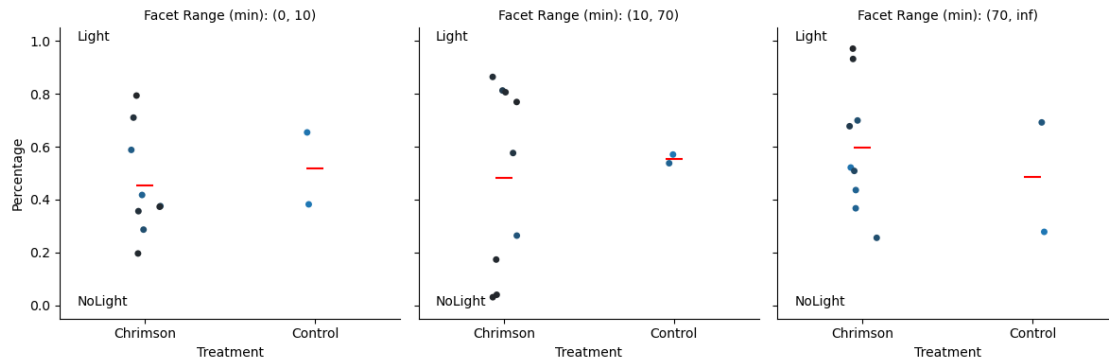
```
##### T-Test #####  
Column = FinalPI, Range Minutes = (0.00 , 10.00)  
Chrimson vs. Control: T=-0.44, p=0.66814
```

```
##### T-Test #####  
Column = FinalPI, Range Minutes = (10.00 , 70.00)  
Chrimson vs. Control: T=-0.30, p=0.77409
```

```
##### T-Test #####
```

```
Column = FinalPI, Range Minutes = (70.00 , inf)
Chrimson vs. Control: T=0.57, p=0.57959
```

```
[8]: arena.plot_percentage_facet(cutoffs=(10,70))
```



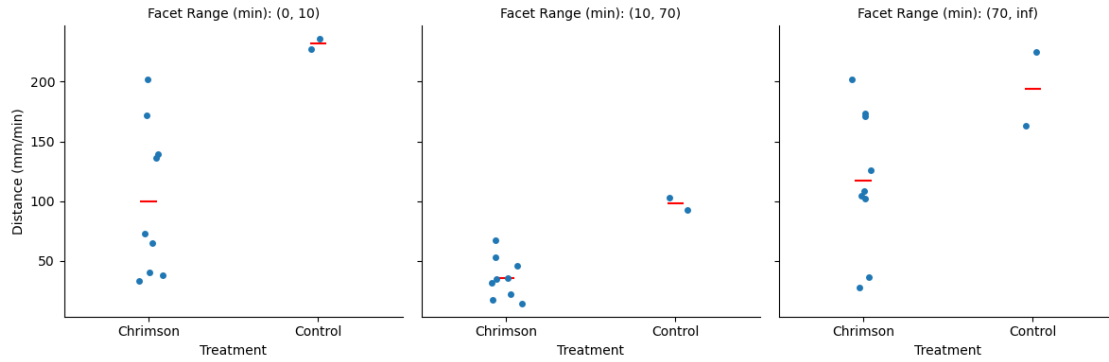
```
[9]: arena.run_pairwise_comparisons_facet(metric="FinalPercentage",cutoffs=(10,70))
```

```
##### T-Test #####
Column = FinalPercentage, Range Minutes = (0.00 , 10.00)
Chrimson vs. Control: T=-0.41, p=0.69405
```

```
##### T-Test #####
Column = FinalPercentage, Range Minutes = (10.00 , 70.00)
Chrimson vs. Control: T=-0.28, p=0.78730
```

```
##### T-Test #####
Column = FinalPercentage, Range Minutes = (70.00 , inf)
Chrimson vs. Control: T=0.57, p=0.58279
```

```
[10]: arena.plot_totaldistance_facet(cutoffs=(10,70))
```



```
[11]: arena.run_pairwise_comparisons_facet(metric="TotalDistance",cutoffs=(10,70))
```

T-Test

Column = TotalDistance, Range Minutes = (0.00 , 10.00)

Chrimson vs. Control: T=-2.81, p=0.02024

T-Test

Column = TotalDistance, Range Minutes = (10.00 , 70.00)

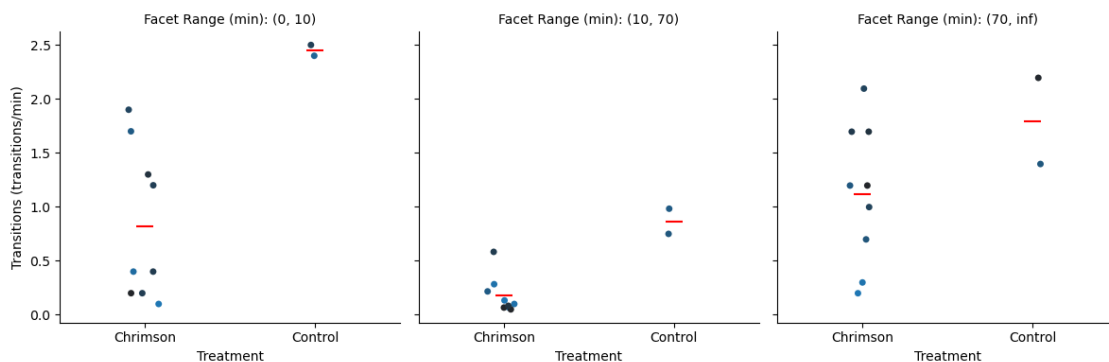
Chrimson vs. Control: T=-4.80, p=0.00097

T-Test

Column = TotalDistance, Range Minutes = (70.00 , inf)

Chrimson vs. Control: T=-1.70, p=0.12271

```
[12]: arena.plot_transitions_facet(cutoffs=(10,70))
```



```
[13]: arena.run_pairwise_comparisons_facet(metric="Transitions",cutoffs=(10,70))
```

```
##### T-Test #####
```

```
Column = Transitions, Range Minutes = (0.00 , 10.00)
```

```
Chrimson vs. Control: T=-3.14, p=0.01195
```

```
##### T-Test #####
```

```
Column = Transitions, Range Minutes = (10.00 , 70.00)
```

```
Chrimson vs. Control: T=-5.18, p=0.00058
```

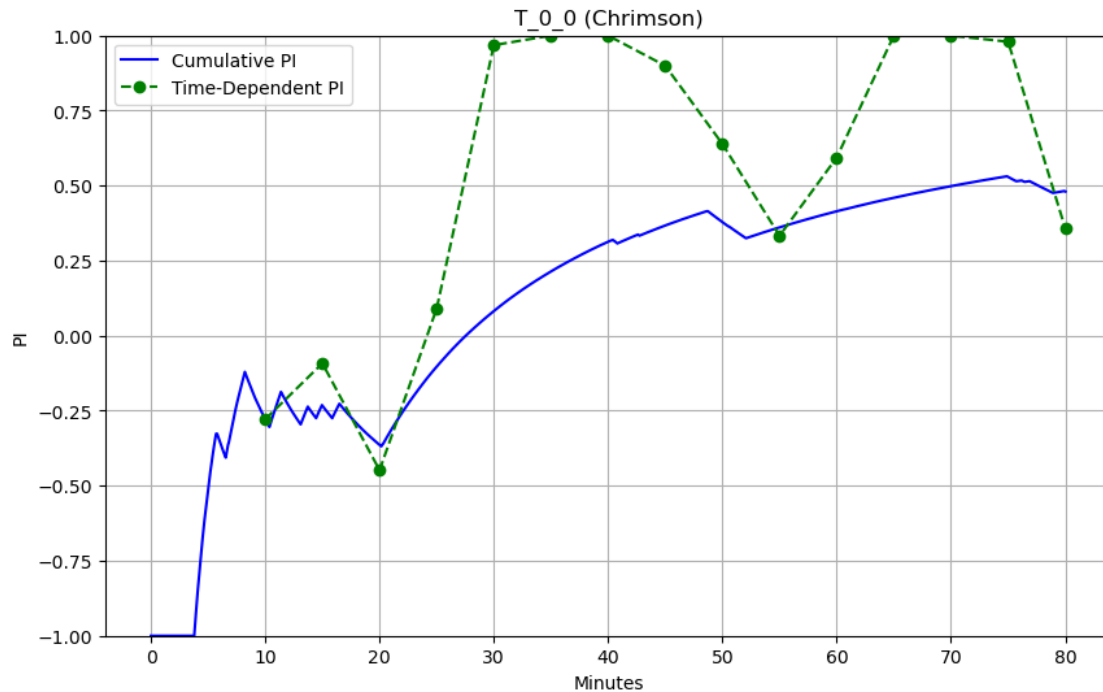
```
##### T-Test #####
```

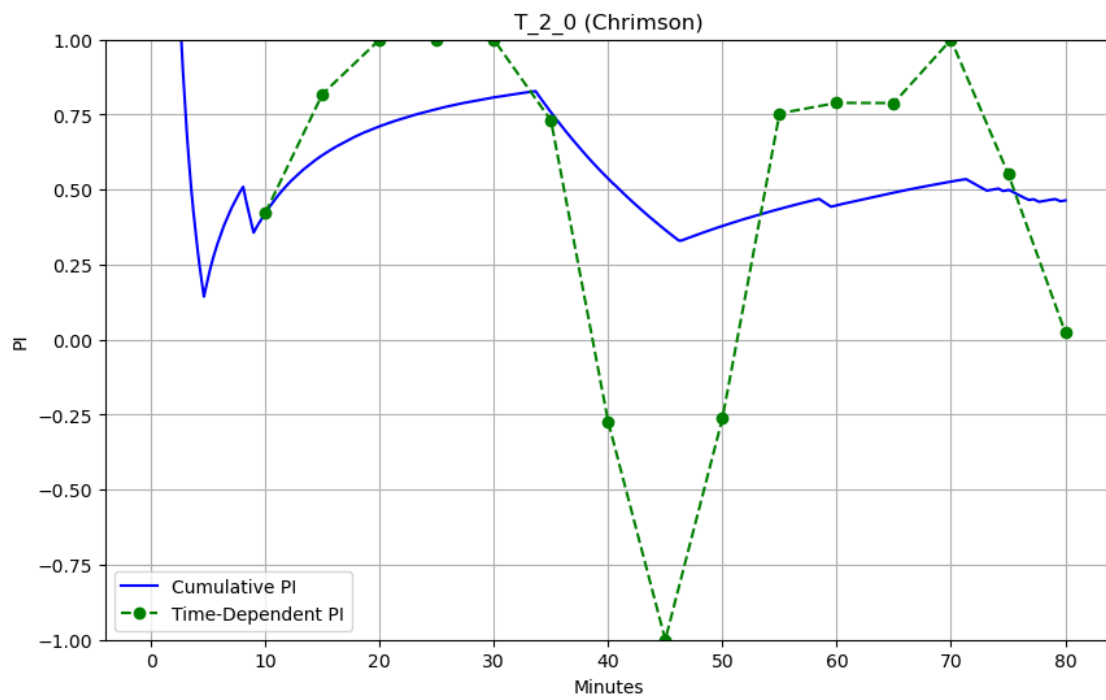
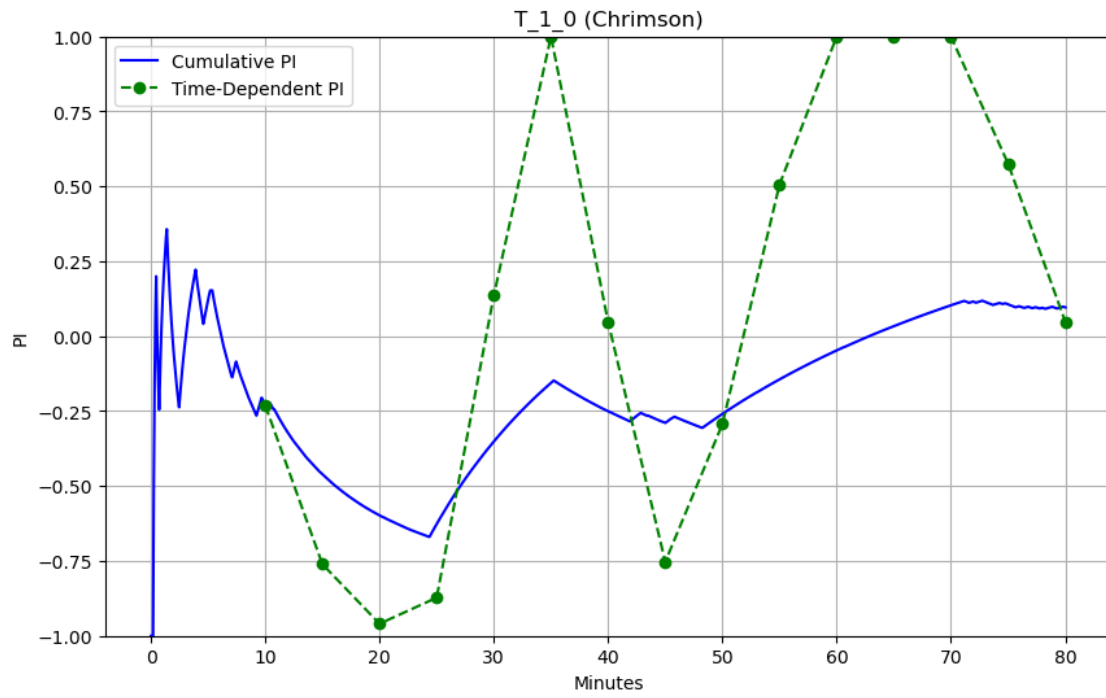
```
Column = Transitions, Range Minutes = (70.00 , inf)
```

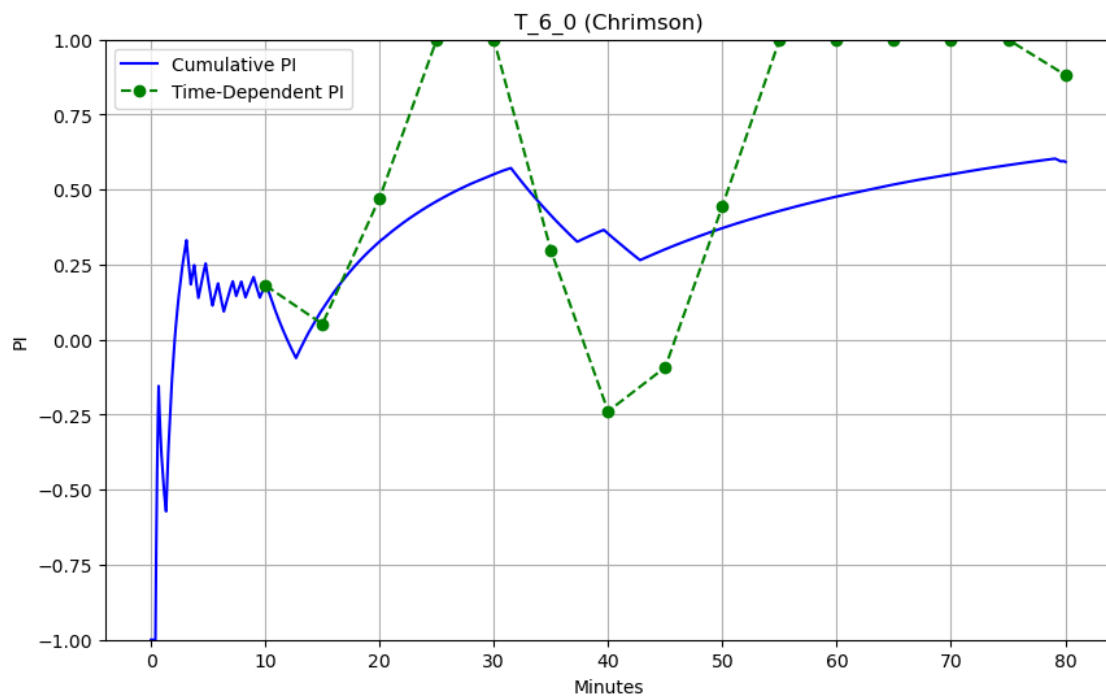
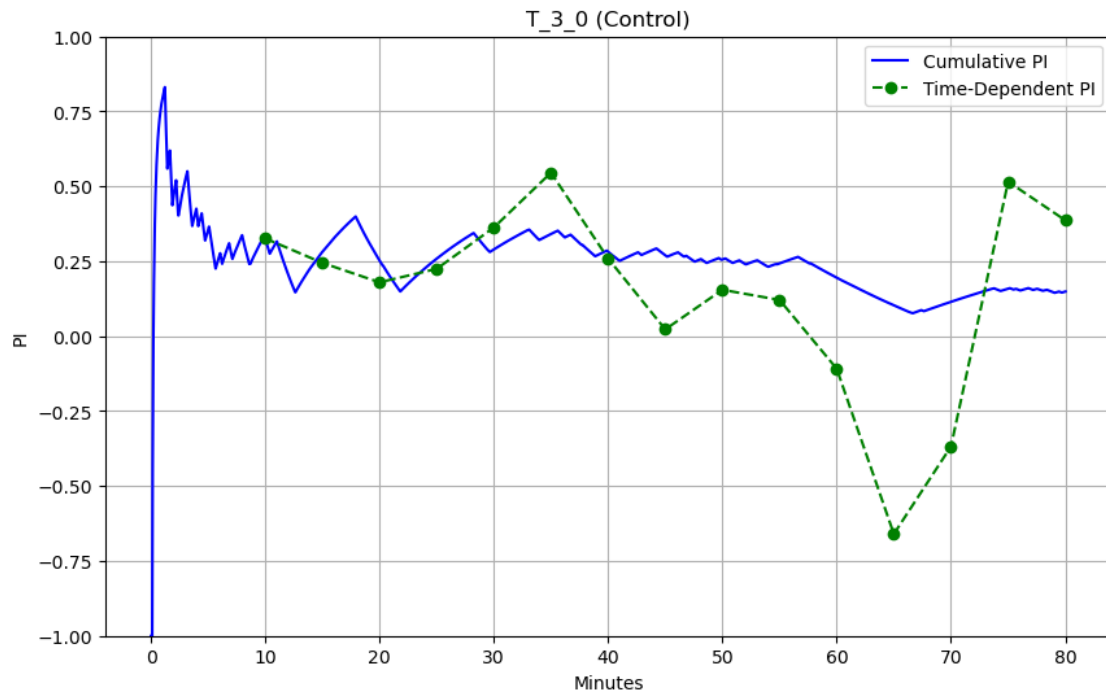
```
Chrimson vs. Control: T=-1.36, p=0.20792
```

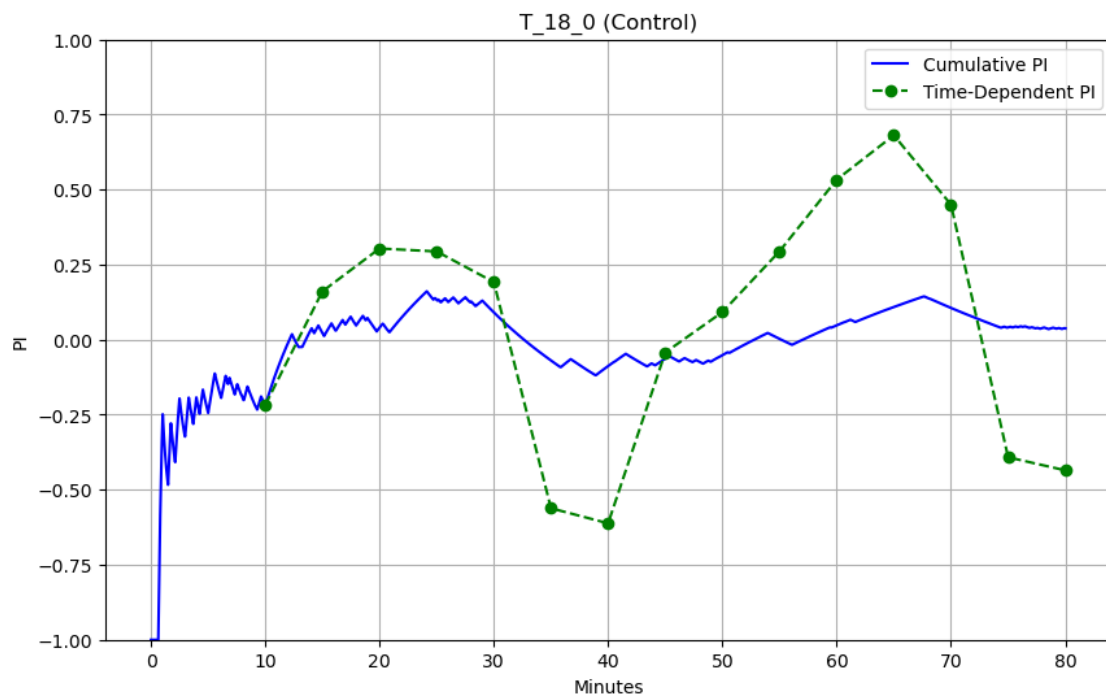
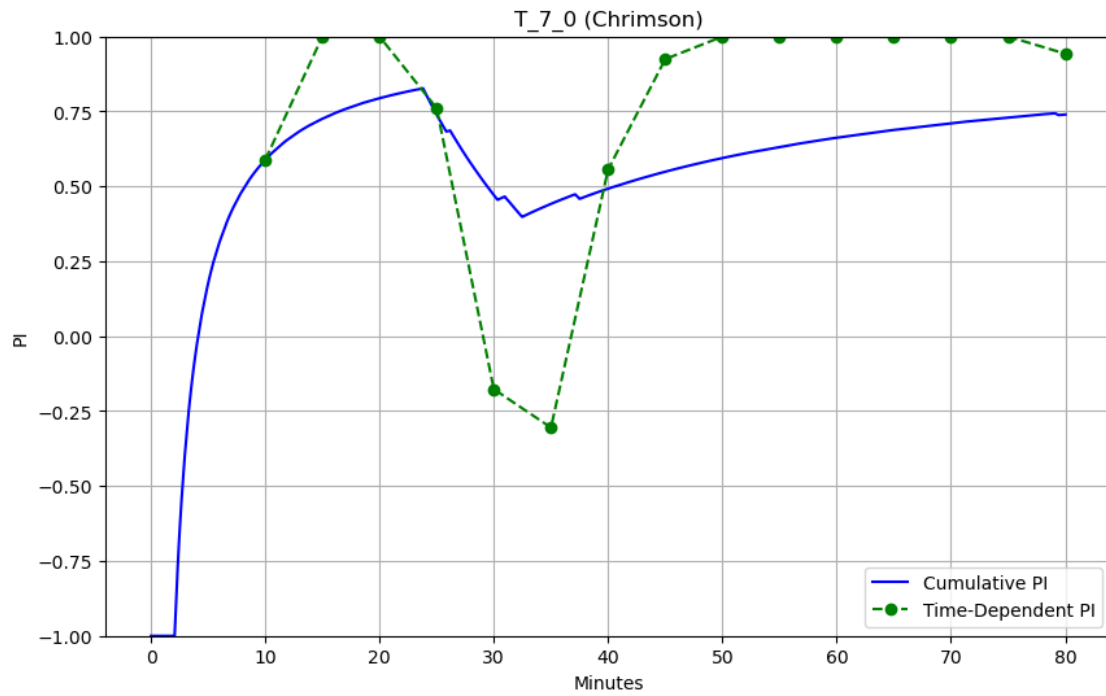
Tracker plots: These show, for each tracker individually, the progression of PI and Percentage showing cumulative values and time-dependent values

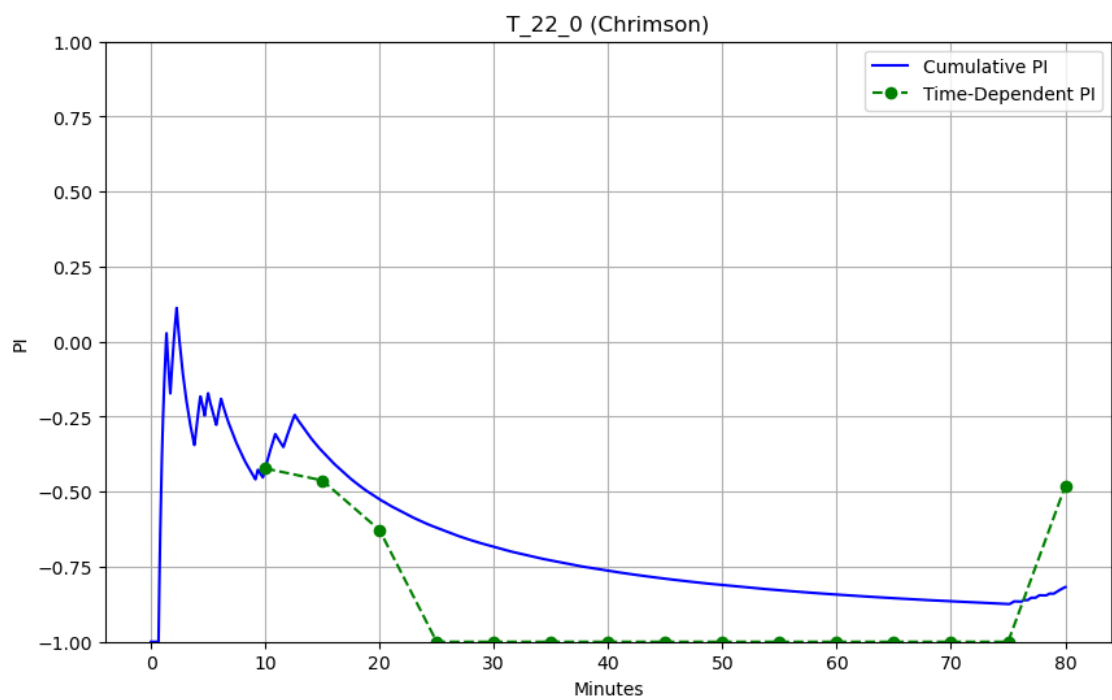
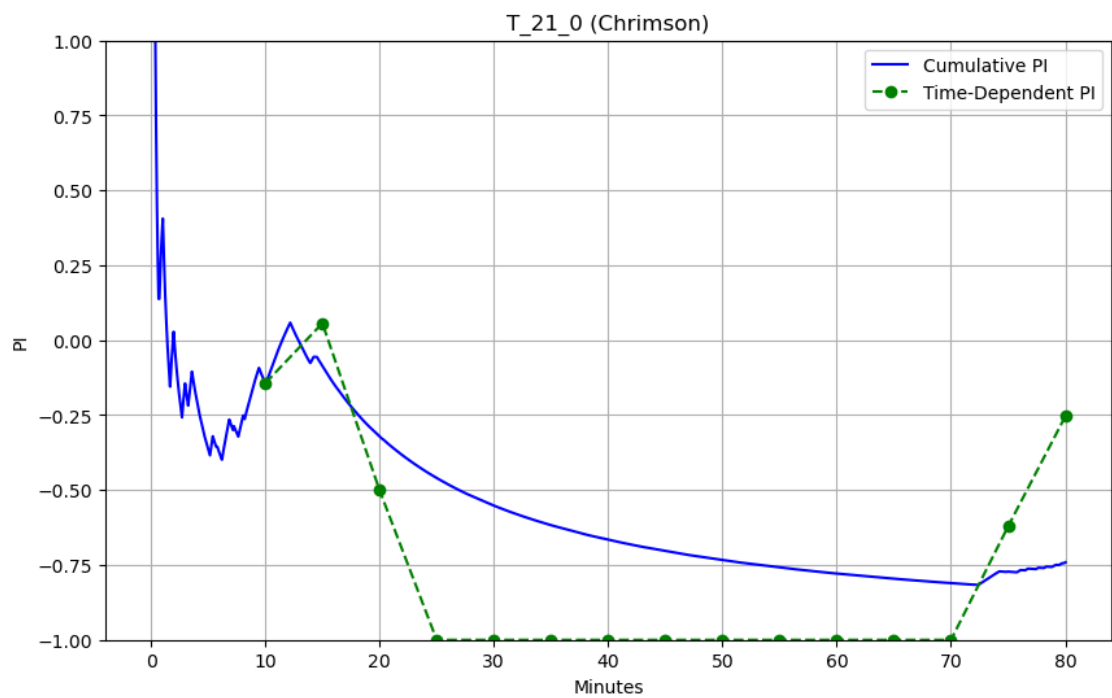
```
[14]: arena.plot_trackers_pis()
```

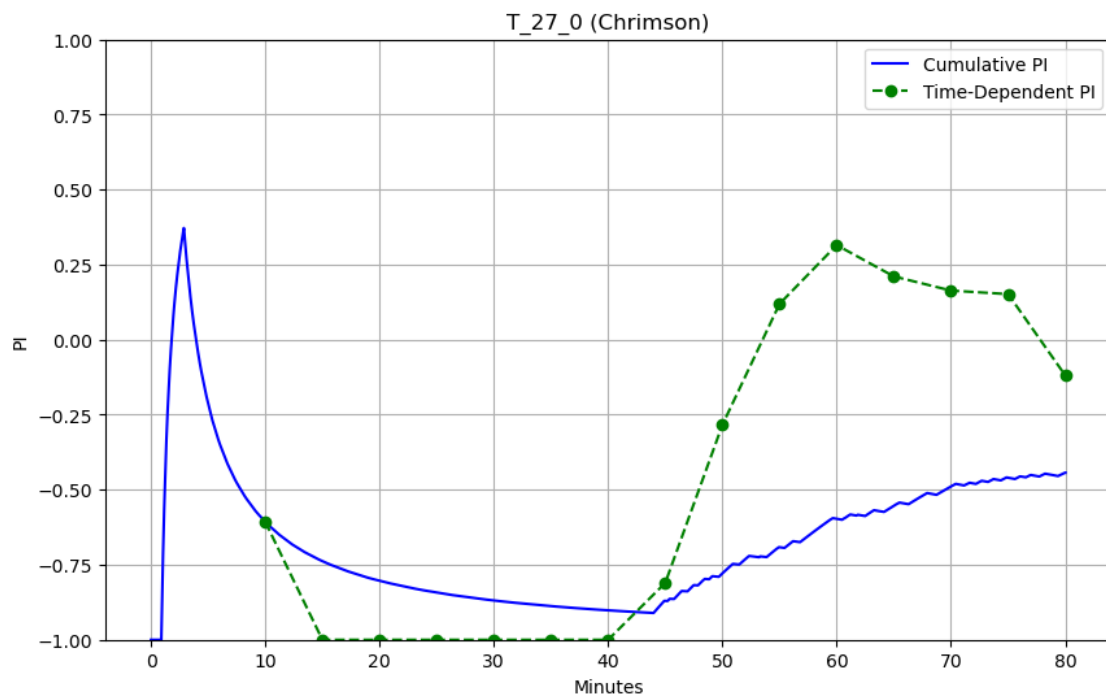
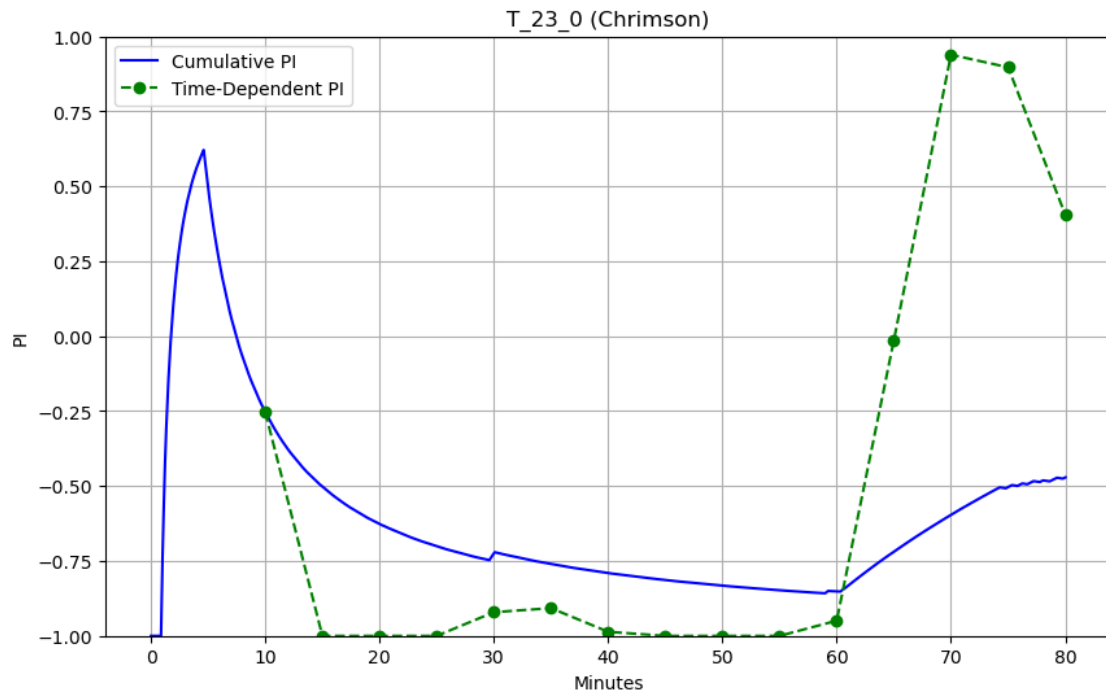






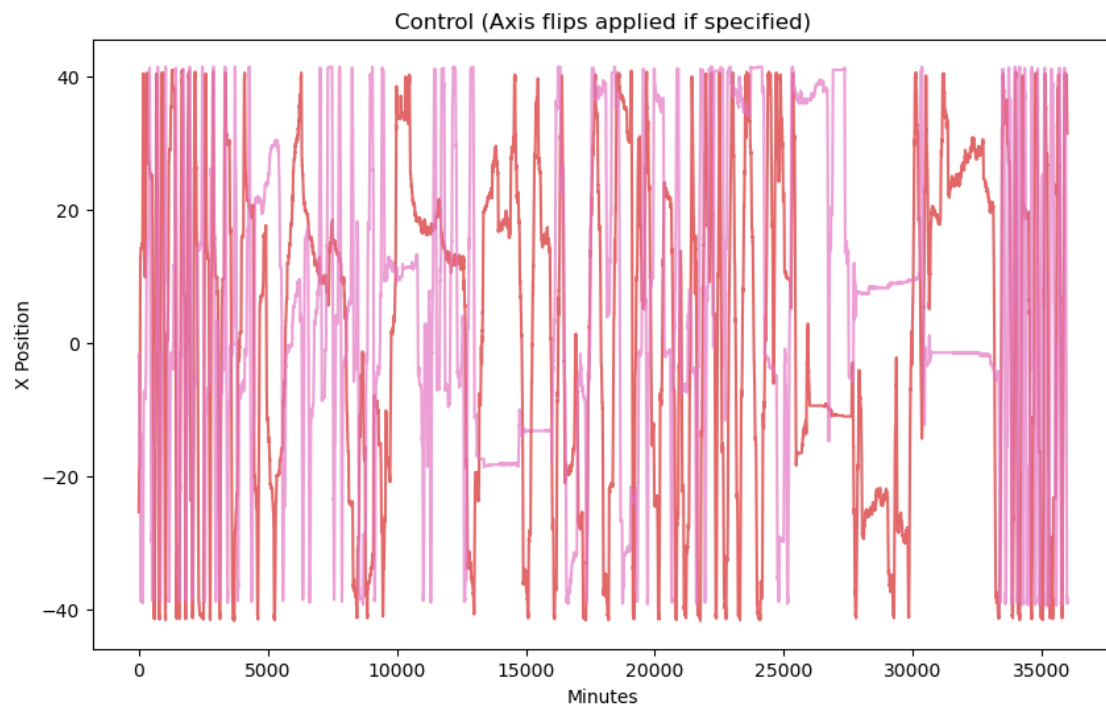
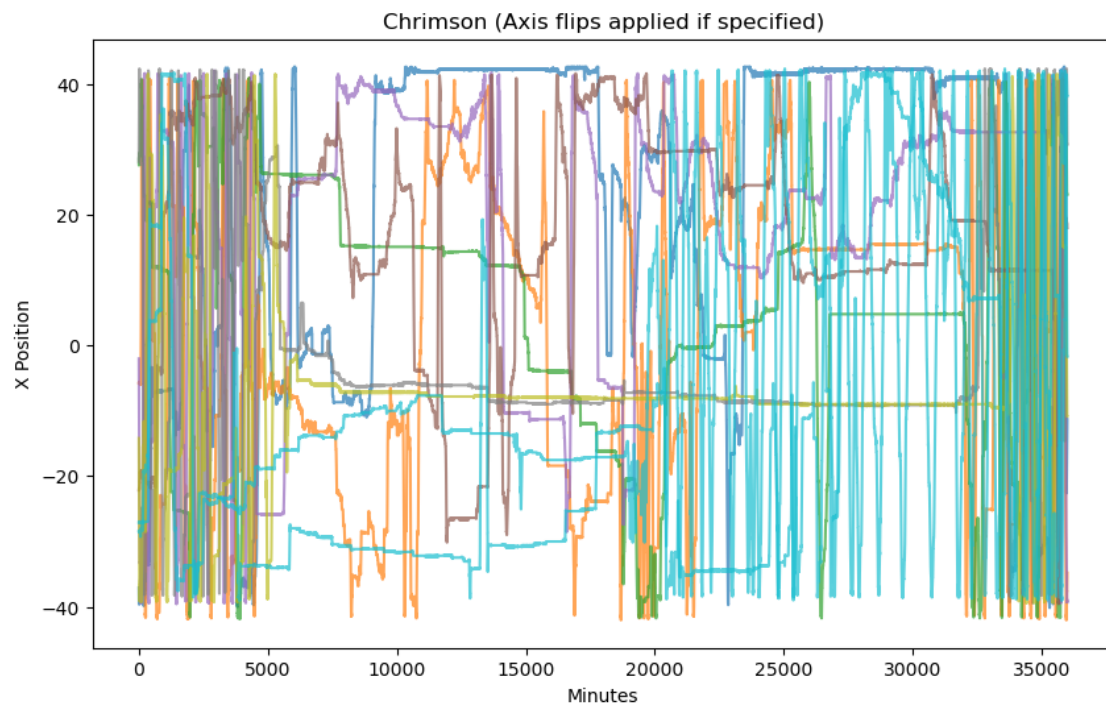






Positional Plots: Show the X positions of each tracker as a function of time.

```
[15]: arena.plot_trackers_x(one_plot=True)
```



```
[16]: arena.plot_trackers_x()
```

